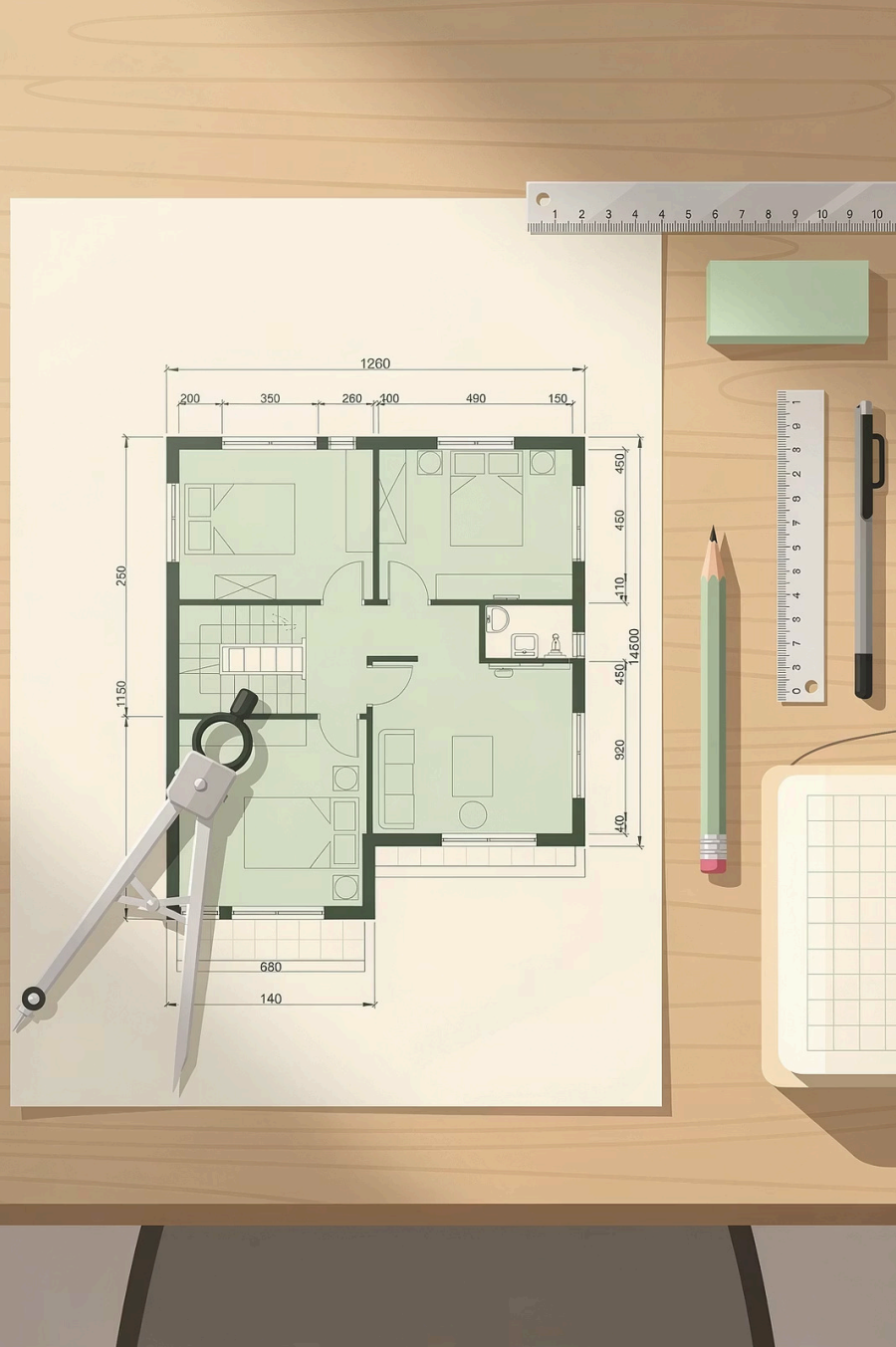




UNLOCK

WEEK 35 · CAPSTONE PLANNING

The Project Charter, KPI Trees & Risk Planning

Laying the strategic foundation for your capstone project — from formal authorization to measurable success and proactive risk management.

Today's Roadmap

This session is dedicated entirely to equipping you with the planning frameworks that separate successful projects from those that drift. We'll move through two major planning instruments, each building on the last.

01	02	03
The Project Charter Understand how to formally authorize and scope your capstone project with a single, powerful document.	KPI Trees Learn to decompose high-level business goals into a hierarchy of measurable Key Performance Indicators.	Risk Planning Identify, assess, and create mitigation strategies for potential threats to your project's success.



UNLOX · SECTION 35.1

What Is a Project Charter?

A **Project Charter** is a high-level document that formally **authorizes** a project and provides a concise, shared definition of its scope, objectives, stakeholders, and constraints. Think of it as the project's constitution — it establishes legitimacy, grants the project manager authority to allocate resources, and aligns every stakeholder around a common purpose before a single task is executed.

Without a charter, projects are vulnerable to scope creep, misaligned expectations, and resource conflicts. A well-crafted charter prevents these problems at the source.

Why the Charter Matters

The charter isn't bureaucratic overhead — it's the single document your sponsor, team, and stakeholders will reference whenever decisions or disputes arise. Here's what it accomplishes:



Alignment

Forces early consensus on purpose, scope, and success criteria among all stakeholders.



Authority

Formally empowers the project manager to commit resources and make key decisions.



Boundaries

Defines what is — and critically, what is *not* — included in the project scope.



Reference Point

Serves as the baseline document to evaluate change requests and resolve conflicts.

Anatomy of a Strong Charter

Every effective project charter contains a consistent set of elements. While formats vary across organizations, the core components remain the same. Use this as your checklist when drafting your capstone charter.

Core Elements

- **Project Title & Description** — A clear, concise statement of what the project is
- **Business Case** — Why this project matters and the problem it solves
- **Objectives & Success Criteria** — SMART goals that define "done"
- **Scope Statement** — In-scope and out-of-scope boundaries

Supporting Elements

- **Key Stakeholders** — Sponsor, team, advisors, and end users
- **High-Level Timeline** — Major milestones and target dates
- **Budget & Resources** — Estimated cost and resource needs
- **Assumptions & Constraints** — Known limitations and working hypotheses
- **Risks (Preliminary)** — Top-level threats identified early

Charter Best Practices

Writing a charter is both an art and a discipline. Keep these principles in mind as you draft your capstone document:

1

Keep It Concise

A charter is typically **1–3 pages**. It's a summary, not a project plan. If you're writing more than three pages, you're likely including execution-level detail that belongs elsewhere.

2

Write SMART Objectives

Every objective should be **Specific, Measurable, Achievable, Relevant, and Time-bound**. Vague goals like "improve the system" will undermine your entire project.

3

Get Sponsor Sign-Off

The charter is only as powerful as the authority behind it. Secure formal approval from your project sponsor before moving to execution.

4

Revisit, Don't Rewrite

Treat the charter as a living reference. You may refine it as clarity improves, but wholesale rewrites signal that planning was premature.

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Introducing the KPI Tree

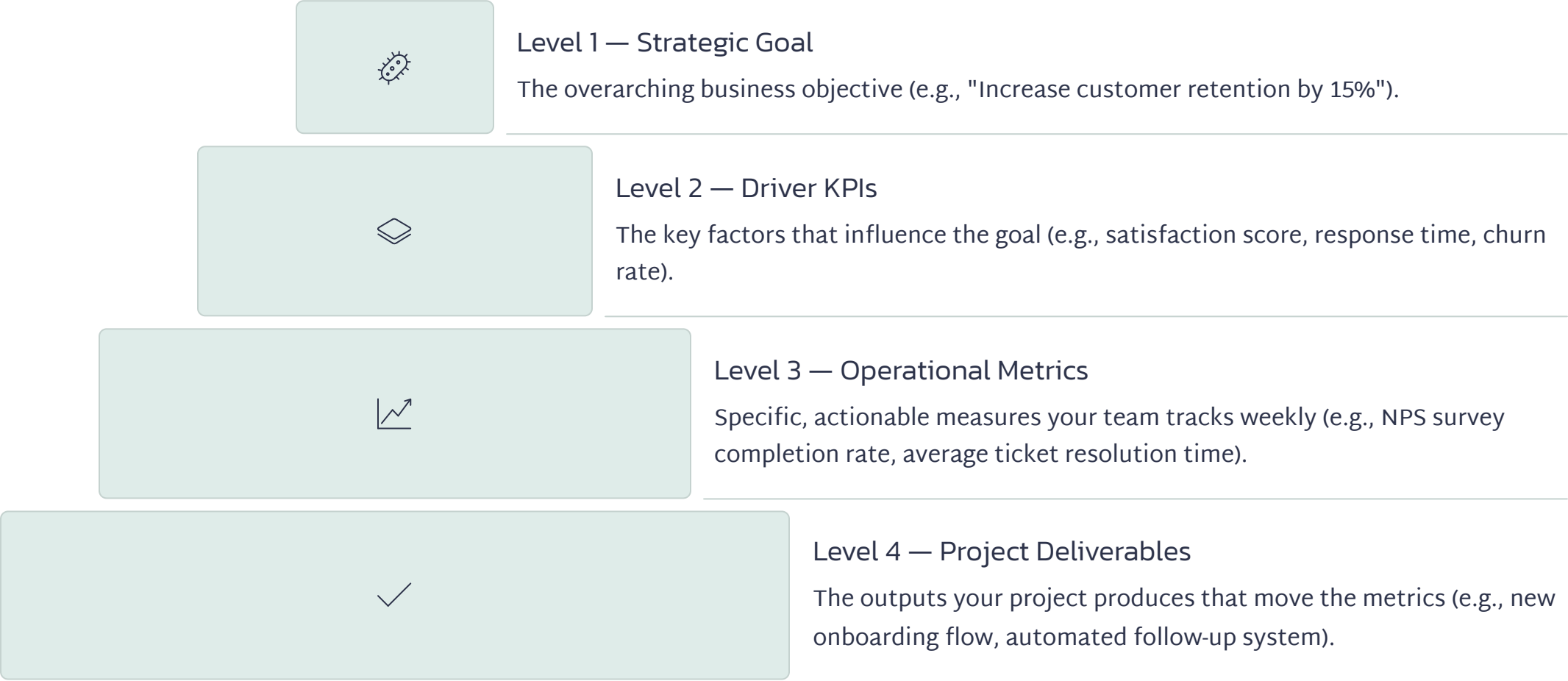
A **KPI Tree** is a structured visualization that decomposes a single high-level business goal into a hierarchy of increasingly specific, **measurable Key Performance Indicators**. It answers the critical question: *"How will we know this project succeeded?"*

By mapping your capstone objectives to quantifiable metrics, the KPI Tree creates a direct, traceable link between daily project activities and the strategic business outcome you're trying to influence. Without this connection, even well-executed projects can fail to demonstrate value.



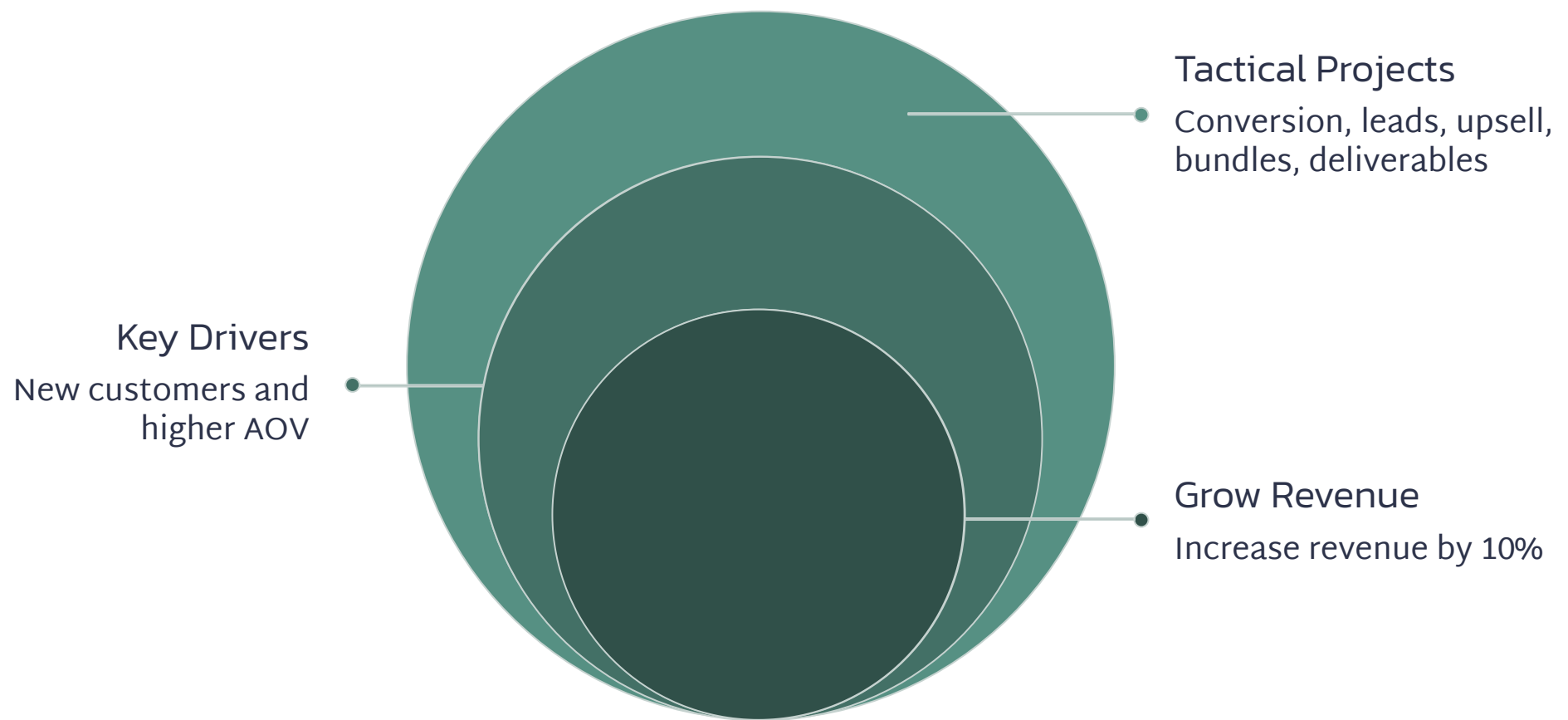
Building Your KPI Tree

Constructing a KPI Tree follows a top-down decomposition process. Start with the broadest goal and progressively break it down until you reach metrics your team can directly influence and measure.



KPI Tree — Example Breakdown

Let's walk through a concrete example. Suppose your capstone project supports a company's goal to **grow quarterly revenue by 10%**. Here's how the KPI Tree might decompose:



Notice how each level gets more specific and actionable. Your capstone deliverables sit at the bottom, directly tied to metrics that roll up to the strategic goal. This traceability is what makes your project's impact defensible.

Common KPI Tree Pitfalls

Even experienced project managers make mistakes when building KPI Trees. Watch for these traps as you construct yours:

Vanity Metrics

Metrics that look impressive but don't connect to business outcomes — like page views without conversion context. Every KPI must have a clear "so what?"

Too Many Levels

More than four levels of decomposition usually means you're over-engineering the tree. Keep it focused and manageable for your team.

Missing Baselines

A KPI without a current baseline is unmeasurable. Before setting targets, document where you're starting from — otherwise, progress is impossible to prove.

Broken Linkages

If a lower-level metric doesn't logically influence the level above it, the tree has a gap. Test each branch by asking: "If this metric improves, does the parent improve?"

Risk Planning — Thinking Ahead

Risk planning is the disciplined process of identifying potential threats to your project and developing strategies to prevent or minimize their impact. It is not pessimism — it is **professional preparedness**. The most successful project managers are those who anticipate problems before they arrive.

For your capstone, risk planning transforms uncertainty from a source of anxiety into a managed variable. Every project carries risk; the difference between success and failure is whether those risks were addressed proactively or discovered too late.



The Risk Management Process

Effective risk management follows a repeatable four-step cycle. This isn't a one-time exercise — revisit your risk register at every major milestone throughout your capstone.



Identify

Brainstorm all possible risks using categories: technical, resource, schedule, stakeholder, and external factors.



Assess

Rate each risk by **probability** (likelihood of occurring) and **impact** (severity if it does). Use a simple High/Medium/Low matrix.



Mitigate

For each high-priority risk, choose a strategy: **avoid**, **transfer**, **reduce**, or **accept**. Document the action plan.



Monitor

Track risk triggers throughout execution. Update the register as new risks emerge and old risks are resolved.

Risk Assessment Matrix

Use a **Probability × Impact matrix** to prioritize your risks. This simple tool helps your team focus mitigation efforts where they matter most. Here's a typical example for a capstone project:

Risk	Probability	Impact	Priority
Key team member drops the course	Medium	High	● High
Data source becomes unavailable	Low	High	● Medium
Scope creep from stakeholder requests	High	Medium	● High
Technology integration issues	Medium	Medium	● Medium
Timeline slippage near final deadline	High	High	● Critical

Risks rated **High** or **Critical** require documented mitigation plans with assigned owners and trigger dates. Medium risks should be monitored at each sprint review.

Four Risk Response Strategies

Once you've assessed your risks, choose the appropriate response strategy for each. Not every risk needs the same treatment — match the strategy to the situation.



Avoid

Eliminate the risk entirely by changing the project plan.
Example: switching to a more reliable data source before work begins.



Transfer

Shift the risk to a third party. Example: using a managed cloud service instead of hosting your own infrastructure.



Reduce

Take actions to lower the probability or impact. Example: building in schedule buffers or cross-training team members on critical tasks.



Accept

Acknowledge the risk and prepare a contingency if it occurs. Best for low-impact risks where mitigation costs exceed potential damage.

Connecting It All Together

The Charter, KPI Tree, and Risk Plan aren't isolated documents — they form an integrated planning system. Here's how each component reinforces the others:



A project with a charter but no KPIs can't prove success. A project with KPIs but no risk plan is one surprise away from failure. Use all three.

Key Takeaways & Next Steps

Draft Your Charter First

Before any execution begins, get alignment on purpose, scope, and authority. Aim for 1–3 pages with SMART objectives and sponsor sign-off.

Build a KPI Tree Early

Decompose your strategic goal into measurable, actionable metrics. Ensure every project deliverable traces upward to a business outcome.

Plan for Risk Continuously

Identify, assess, and mitigate risks now — and revisit them at every milestone. Proactive teams recover faster and deliver more reliably.

Your assignment: Submit a first draft of your capstone Project Charter, a preliminary KPI Tree, and a risk register with at least five identified risks and their response strategies by next session.